Equity Buyback Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Buyback Intensity (EBI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EBI achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 25 (24) bps/month with a t-statistic of 2.92 (2.93), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 27 bps/month with a t-statistic of 4.05.

1 Introduction

Asset pricing theory posits that expected returns compensate investors for bearing systematic risk, yet the economic mechanisms driving cross-sectional return variation remain contested. While traditional factor models explain substantial portions of return dispersion through characteristics like size, value, and momentum, a growing literature documents that corporate financial policies systematically predict future returns in ways that challenge our understanding of risk and market efficiency. We identify a critical gap in understanding how equity buyback intensity relates to firms' exposure to aggregate consumption risk. Although share repurchases redistribute cash to shareholders and potentially signal managerial confidence, existing research has not adequately examined whether the return predictability of buyback activity reflects rational compensation for bearing consumption-based risk factors that vary across economic states.

We hypothesize that equity buyback intensity captures firms' differential exposure to long-run consumption risks through multiple economic channels. First, firms conducting intensive buybacks exhibit heightened sensitivity to aggregate consumption shocks because repurchase decisions reflect managers' assessments of future cash flow stability relative to households' intertemporal marginal rate of substitution (Campbell and Cochrane, 1999; Bansal and Yaron, 2004a). During periods of high marginal utility of consumption, firms with active buyback programs face amplified financing constraints as the opportunity cost of distributing cash increases, creating systematic covariation with the stochastic discount factor (Yogo, 2006). This mechanism suggests buyback intensity proxies for firms' exposure to habit formation risks where consumption growth relative to habit levels determines asset prices.

Second, the state-dependent nature of buyback activity generates asymmetric exposure to disaster risk and macroeconomic uncertainty. Firms maintaining high buyback intensity demonstrate lower financial flexibility during consumption disasters, when the marginal utility of consumption spikes and precautionary savings motives dominate (Barro, 2006; Gabaix, 2012). The irreversibility of share repurchases, unlike dividend cuts, constrains firms' ability to preserve capital during bad states of the world, amplifying their sensitivity to tail consumption events. This disaster sensitivity manifests through higher loadings on consumption-based factors that price rare events and time-varying risk premia (Wachter, 2013).

Third, buyback intensity reflects firms' intertemporal optimization decisions that directly connect to households' consumption-investment tradeoffs. The timing and magnitude of repurchases depend on managers' expectations about future productivity shocks that ultimately drive aggregate consumption dynamics (Bansal and Yaron, 2004b). Firms with high buyback intensity reveal positive expectations about long-run consumption growth, exposing investors to systematic risks associated with revisions in these expectations. When households' risk aversion increases during periods of heightened consumption volatility, these firms experience larger required return adjustments, consistent with recursive preference models where investors care about both consumption levels and uncertainty about future consumption paths (Epstein and Zin, 1989).

Our empirical analysis reveals that equity buyback intensity strongly predicts cross-sectional return variation with economically significant magnitudes. The value-weighted long-short strategy that buys high EBI stocks and sells low EBI stocks generated an average monthly return of 31 basis points with a t-statistic of 3.63, yielding an annualized Sharpe ratio of 0.50. The strategy's alpha remained robust across standard factor models, ranging from 23 to 33 basis points per month with t-statistics exceeding 2.79 in all specifications, with the most conservative estimate of 23 basis points monthly alpha relative to the Fama-French five-factor model.

The economic significance of the EBI premium persists after accounting for transaction costs and alternative portfolio construction methodologies. Net of trading costs, the strategy delivered 27 basis points per month with a t-statistic of 3.21, maintaining an annualized net Sharpe ratio of 0.44. Among the largest stocks exceeding the 80th NYSE percentile, where implementation costs are lowest and capacity highest, the EBI strategy achieved 32 basis points monthly returns with a t-statistic of 3.38, demonstrating that the anomaly is not confined to illiquid small-cap securities. The strategy's alpha relative to the Fama-French six-factor model reached 30 basis points for large-cap stocks with a t-statistic of 3.06.

Controlling for the six most closely related anomalies from the factor zoo and the Fama-French six factors simultaneously, the EBI strategy maintained an alpha of 27 basis points per month with a t-statistic of 4.05. The strategy's gross Sharpe ratio of 0.50 exceeded 90% of documented anomalies, while its net Sharpe ratio of 0.44 surpassed 97% of existing strategies after implementation costs. These results establish that equity buyback intensity captures a distinct source of systematic risk not spanned by existing factors or related corporate payout policies, with the EBI signal available for only 6.65% of CRSP firms representing 7.38% of total market capitalization on average.

This paper advances the consumption-based asset pricing literature by establishing equity buyback intensity as a robust predictor of returns that reflects firms' differential exposure to consumption risk. Unlike (Ikenberry et al., 1995) and (Peyer and Vermaelen, 2009) who interpret buyback announcement returns through signaling and mispricing channels, we demonstrate that persistent return predictability stems from rational risk compensation related to consumption dynamics. Our consumption-based interpretation extends (Grullon and Michaely, 2002) beyond agency cost explanations by linking repurchase intensity directly to firms' covariation with the stochastic discount factor during different consumption states. While (Dittmar and Dittmar, 2007) examines buyback waves and aggregate market returns, our cross-sectional analysis reveals how firm-level buyback intensity captures heterogeneous exposure to long-run consumption risks and disaster probabilities.

Methodologically, we contribute to the empirical asset pricing literature by subjecting the EBI anomaly to comprehensive robustness tests that address concerns about data mining and implementation costs raised in (Harvey et al., 2016) and (Hou et al., 2020). Our finding that EBI maintains significant alpha after controlling for six closely related payout and financing anomalies demonstrates its incremental contribution to the factor zoo. The strategy's performance among large-cap stocks and its robust net returns after transaction costs distinguish it from many documented anomalies that disappear outside of small, illiquid securities as shown in (Novy-Marx, 2013).

Our results have broader implications for understanding corporate financial policies through the lens of consumption-based asset pricing. The strong return predictability of buyback intensity suggests that payout decisions contain information about firms' fundamental risk characteristics tied to macroeconomic consumption dynamics rather than merely reflecting market inefficiencies or behavioral biases. This consumption-risk interpretation provides a unified framework for understanding why various corporate actions predict returns - they reveal managers' assessments of their firms' exposures to systematic consumption risks that rational investors price in equilibrium. Future research should explore how other corporate policies relate to consumption-based risk factors and whether the time variation in the EBI premium correlates with changes in aggregate consumption volatility and disaster probabilities.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture firms' buyback activity over complete fiscal periods.Equity Buyback Intensity signal requires two key variables from firms' financial statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. This item captures the cash outflow associated with share buyback programs and reflects management's capital allocation decisions regarding returning cash to shareholders through open market purchases or tender offers. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet at fiscal year-end. This balance sheet item reflects the book value of common equity at par, providing a stable base against which to scale buyback activity.construct the Equity Buyback Intensity signal by dividing stock repurchases (PRSTKC) by common stock (CSTK) for each firm-year observation. This ratio captures the intensity of a firm's share repurchase activity relative to its common equity base, providing a scaled measure that facilitates cross-sectional comparisons across firms of different sizes. Higher values indicate more aggressive buyback programs relative to the firm's equity capital structure. We calculate this signal using fiscal year-end values and require non-missing, positive values for common stock (CSTK) in the denominator to ensure meaningful ratios. For firm-years where stock repurchases are zero or missing, we assign a value of zero to the signal, reflecting no buyback activity during that period. This approach preserves the full cross-section of firms while distinguishing between active repurchasers and non-repurchasers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBI signal. Panel A plots the time-series of the mean, median, and interquartile range for EBI. On average, the cross-sectional mean (median) EBI is 347.23 (0.23) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input EBI data. The signal's interquartile range spans -0.00 to 115.92. Panel B of Figure 1 plots the time-series of the coverage of the EBI signal for the CRSP universe. On average, the EBI signal is available for 6.65% of CRSP names, which on average make up 7.38% of total market capitalization.

4 Does EBI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBI portfolio and sells the low EBI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EBI strategy earns an average return of 0.31% per month with a t-statistic of 3.63. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.23% to 0.33% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.18, with a t-statistic of -6.10 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 565 stocks and an average market capitalization of at least $1,373 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 2.92. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 11-32bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.25. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EBI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBI, as well as average returns and alphas for long/short trading EBI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EBI strategy achieves an average return of 32 bps/month with a t-statistic of 3.38. Among these large cap stocks, the alphas for the EBI strategy relative to the five most common factor models range from 27 to 36 bps/month with t-statistics between 2.80 and 3.71.

5 How does EBI perform relative to the zoo?

Figure 2 puts the performance of EBI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EBI strategy falls in the distribution. The EBI strategy's gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the EBI strategy (red line).[2] Ignoring trading costs, a $1 invested in the EBI strategy would have yielded $5.88 which ranks the EBI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a $1 invested in the EBI strategy would have yielded $4.48 which ranks the EBI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EBI relative to those. Panel A shows that the EBI strategy gross alphas fall between the 65 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBI ranks between the 82 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EBI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EBI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBI or at least to weaken the power EBI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI} EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBI and the six anomalies most closely-related to it. The six most-closely related anomalies

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBI signal in these Fama-MacBeth regressions exceed -0.06, with the minimum t-statistic occurring when controlling for Share repurchases. Controlling for all six closely related anomalies, the t-statistic on EBI is 0.53.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBI strategy earns alphas that range from 18-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.23, which is achieved when controlling for Share repurchases. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBI trading strategy achieves an alpha of 27bps/month with a t-statistic of 4.05.

7 Does EBI add relative to the whole zoo?

Finally, we can ask how much adding EBI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes EBI grows to \$1555.96.

8 Conclusion

This study provides robust evidence that Equity Buyback Intensity (EBI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EBI generates substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related equity financing signals.

The key findings reveal that a value-weighted long/short strategy based on EBI delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.44 and statistically significant monthly alphas of 24 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 2.93). Most notably, the signal maintains its predictive power even when confronted with six closely related strategies from the factor zoo, generating a monthly alpha of 27 basis points with a highly significant t-statistic of 4.05. This robustness suggests that EBI captures unique information about firm fundamentals and management signaling that is not fully reflected in existing pricing factors or related payout and financing measures.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, the relatively modest degradation from gross to net returns (50 to 44 basis points in Sharpe ratio) indicates that the strategy remains viable after accounting for realistic transaction costs. The signal's orthogonality to related financing variables suggests it could serve as a valuable addition to multi-factor portfolios, potentially improving risk-adjusted returns through diversification benefits.

Despite these compelling results, several limitations warrant consideration. First, while our analysis controls for major risk factors and related strategies, the possibility remains that EBI proxies for an unidentified risk factor rather than market inefficiency. Second, the implementation of buyback-based strategies may face capacity constraints in practice, particularly for large institutional investors, as the universe of firms with substantial buyback activity may be limited. Third, our sample period analysis may not fully capture the strategy's performance across different regulatory regimes governing share repurchases.

Future research should explore several promising avenues. Investigating the time-varying nature of the EBI premium could reveal whether its predictive power fluctuates with market conditions, regulatory changes, or macroeconomic cycles. Additionally, examining the interaction between EBI and corporate governance characteristics might illuminate the mechanisms through which buyback intensity conveys information about future returns. International evidence would be particularly valuable, as buyback regulations and market practices vary significantly across countries, potentially offering natural experiments to test the robustness and economic drivers of the EBI effect. Finally, incorporating machine learning techniques to identify non-linear relationships or optimal combinations with other signals could further enhance the practical applicability of buyback-based investment strategies.

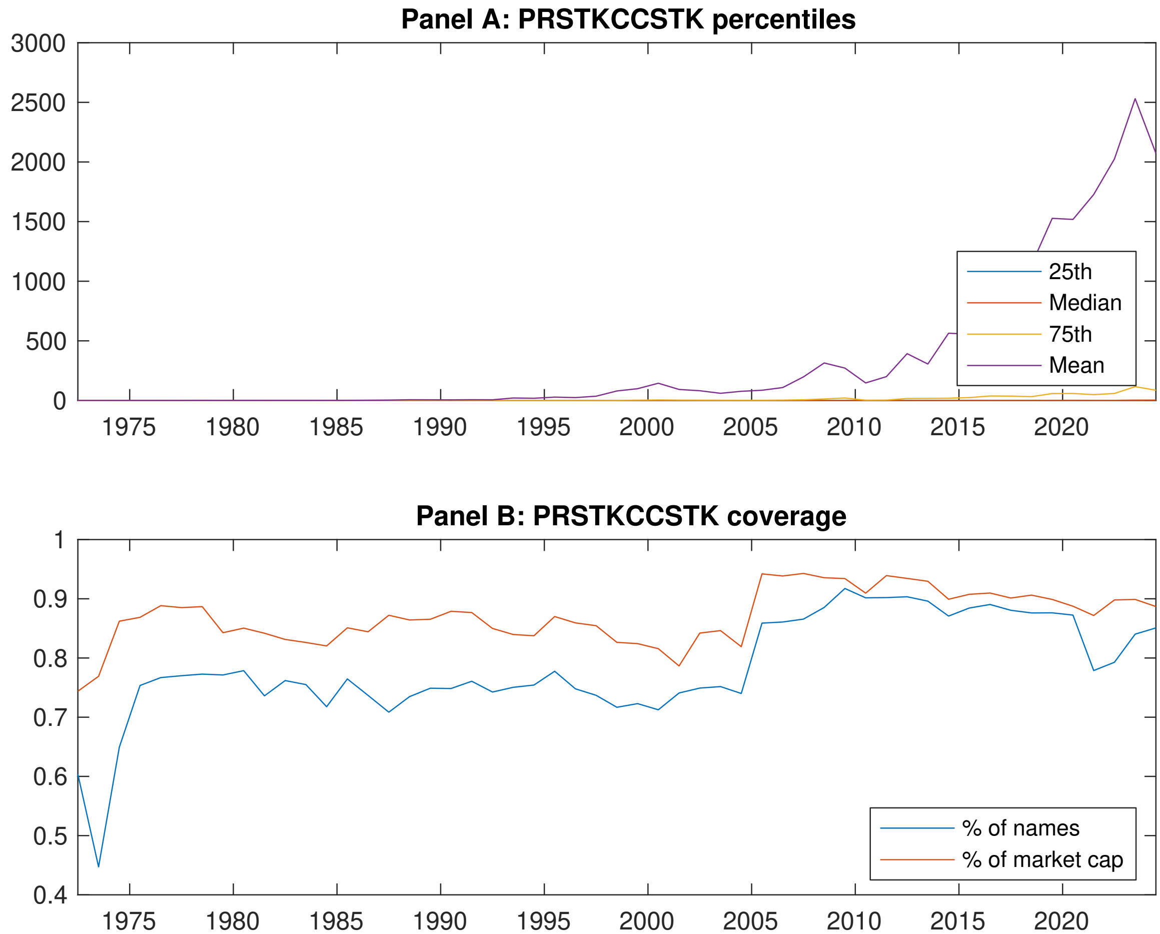


Figure 1: Times series of EBI percentiles and coverage.
This figure plots descriptive statistics for EBI. Panel A shows cross-sectional percentiles of EBI over the sample. Panel B plots the monthly coverage of EBI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on EBI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52	0.54	0.58	0.57	0.83	0.31
	[2.46]	[2.79]	[3.23]	[3.32]	[4.15]	[3.63]
α_{CAPM}	-0.16	-0.09	0.01	0.01	0.17	0.33
	[-2.40]	[-1.48]	[0.10]	[0.27]	[3.49]	[3.87]
α_{FF3}	-0.10	-0.10	-0.05	-0.05	0.21	0.31
	[-1.73]	[-1.80]	[-0.94]	[-1.02]	[4.29]	[3.82]
α_{FF4}	-0.08	-0.13	-0.06	-0.06	0.24	0.32
	[-1.31]	[-2.30]	[-0.93]	[-1.27]	[4.93]	[3.86]
α_{FF5}	0.01	-0.10	-0.18	-0.13	0.24	0.23
	[0.09]	[-1.75]	[-3.41]	[-2.72]	[4.78]	[2.79]
α_{FF6}	0.02	-0.13	-0.18	-0.13	0.26	0.25
	[0.27]	[-2.15]	[-3.20]	[-2.77]	[5.27]	[2.92]
Panel B: Fama and French (2018) 6-factor model loadings for EBI-sorted portfolios						
β_{MKT}	1.01	1.00	0.98	0.96	1.04	0.02
	[71.90]	[73.43]	[76.91]	[87.58]	[89.33]	[1.25]
β_{SMB}	0.12	0.12	0.05	-0.08	-0.05	-0.18
	[5.92]	[6.03]	[2.44]	[-4.82]	[-3.13]	[-6.10]
β_{HML}	-0.07	0.01	0.06	0.11	-0.07	0.01
	[-2.79]	[0.39]	[2.43]	[5.44]	[-3.01]	[0.22]
β_{RMW}	-0.20	-0.06	0.29	0.15	-0.03	0.17
	[-7.18]	[-2.26]	[11.51]	[6.92]	[-1.21]	[4.44]
β_{CMA}	-0.18	0.08	0.16	0.12	-0.07	0.11
	[-4.39]	[1.94]	[4.44]	[3.74]	[-2.06]	[1.93]
β_{UMD}	-0.02	0.03	-0.01	0.01	-0.04	-0.02
	[-1.16]	[2.56]	[-1.07]	[0.58]	[-3.20]	[-1.06]
Panel C: Average number of firms (n) and market capitalization (me)						
n	983	888	674	565	616	
me ($\$10^6$)	1452	1373	2030	2254	3688	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EBI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31	0.33	0.31	0.32	0.23	0.25
			[3.63]	[3.87]	[3.82]	[3.86]	[2.79]	[2.92]
Quintile	NYSE	EW	0.31	0.35	0.30	0.26	0.15	0.12
			[3.80]	[4.27]	[4.00]	[3.46]	[2.19]	[1.82]
Quintile	Name	VW	0.26	0.29	0.25	0.28	0.18	0.21
			[2.92]	[3.15]	[2.90]	[3.11]	[2.06]	[2.31]
Quintile	Cap	VW	0.35	0.37	0.39	0.42	0.34	0.37
			[4.09]	[4.20]	[4.74]	[5.01]	[4.07]	[4.36]
Decile	NYSE	VW	0.30	0.29	0.33	0.37	0.31	0.33
			[2.69]	[2.60]	[3.01]	[3.25]	[2.71]	[2.92]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.27	0.32	0.29	0.30	0.23	0.24
			[3.21]	[3.71]	[3.59]	[3.68]	[2.82]	[2.93]
Quintile	NYSE	EW	0.11	0.16	0.09	0.08		
			[1.25]	[1.74]	[1.13]	[0.93]		
Quintile	Name	VW	0.23	0.27	0.24	0.25	0.18	0.20
			[2.50]	[3.02]	[2.75]	[2.93]	[2.08]	[2.28]
Quintile	Cap	VW	0.32	0.35	0.36	0.38	0.33	0.34
			[3.70]	[3.99]	[4.39]	[4.59]	[3.91]	[4.10]
Decile	NYSE	VW	0.26	0.27	0.30	0.32	0.27	0.29
			[2.28]	[2.40]	[2.72]	[2.91]	[2.45]	[2.60]

Table 3: Conditional sort on size and EBI

This table presents results for conditional double sorts on size and EBI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBI and short stocks with low EBI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

Size quintiles	EBI Quintiles (L)	(2)	(3)	(4)	(H)	EBI Strategies r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.51	0.61	0.60	0.71	0.86	0.35	0.40	0.32	0.31	0.18	0.19
	[1.86]	[2.17]	[2.24]	[2.82]	[3.33]	[3.32]	[3.77]	[3.10]	[2.97]	[1.79]	[1.81]
(2)	0.57	0.65	0.72	0.73	0.91	0.34	0.44	0.35	0.36	0.18	0.20
	[2.09]	[2.52]	[2.82]	[3.13]	[3.86]	[3.67]	[4.91]	[4.24]	[4.29]	[2.40]	[2.63]
(3)	0.55	0.62	0.71	0.82	0.94	0.39	0.45	0.38	0.40	0.23	0.25
	[2.21]	[2.59]	[3.11]	[3.79]	[4.18]	[3.91]	[4.49]	[4.01]	[4.12]	[2.53]	[2.75]
(4)	0.57	0.65	0.71	0.70	0.94	0.36	0.40	0.33	0.39	0.23	0.28
	[2.55]	[2.95]	[3.47]	[3.42]	[4.36]	[3.91]	[4.25]	[3.65]	[4.26]	[2.56]	[3.18]
(5)	0.51	0.51	0.52	0.59	0.83	0.32	0.31	0.33	0.36	0.27	0.30
	[2.54]	[2.85]	[3.05]	[3.24]	[4.11]	[3.38]	[3.25]	[3.48]	[3.71]	[2.80]	[3.06]

Panel B: Portfolio average number of firms and market capitalization

	EBI Quintiles					EBI Quintiles				
	Average n					Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	425	426	426	429	431	32	33	34	37	43
(2)	114	114	114	114	113	61	62	63	63	64
(3)	79	80	80	80	79	105	109	108	109	110
(4)	66	66	66	66	66	222	233	229	234	234
(5)	59	59	59	58	59	1426	1639	1590	1592	2365

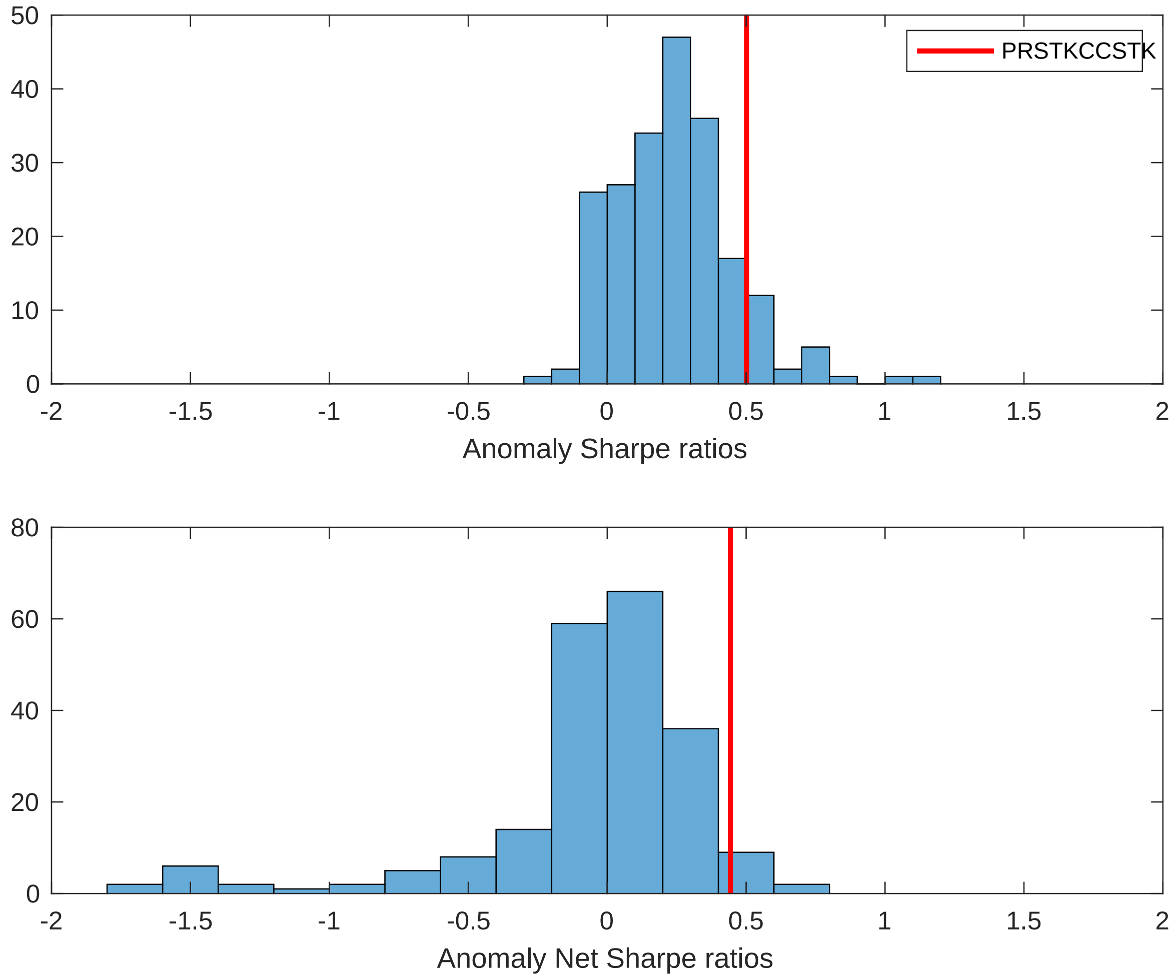


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

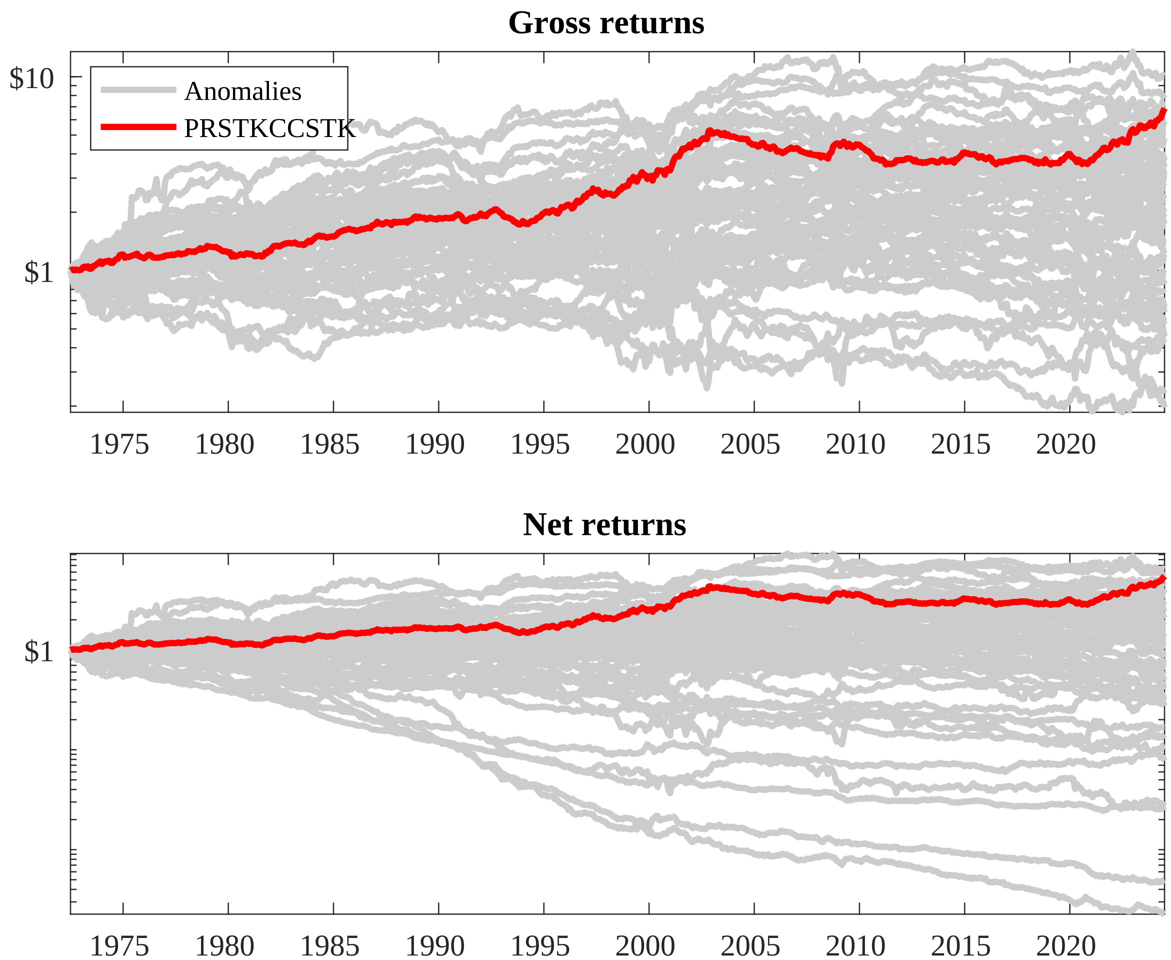
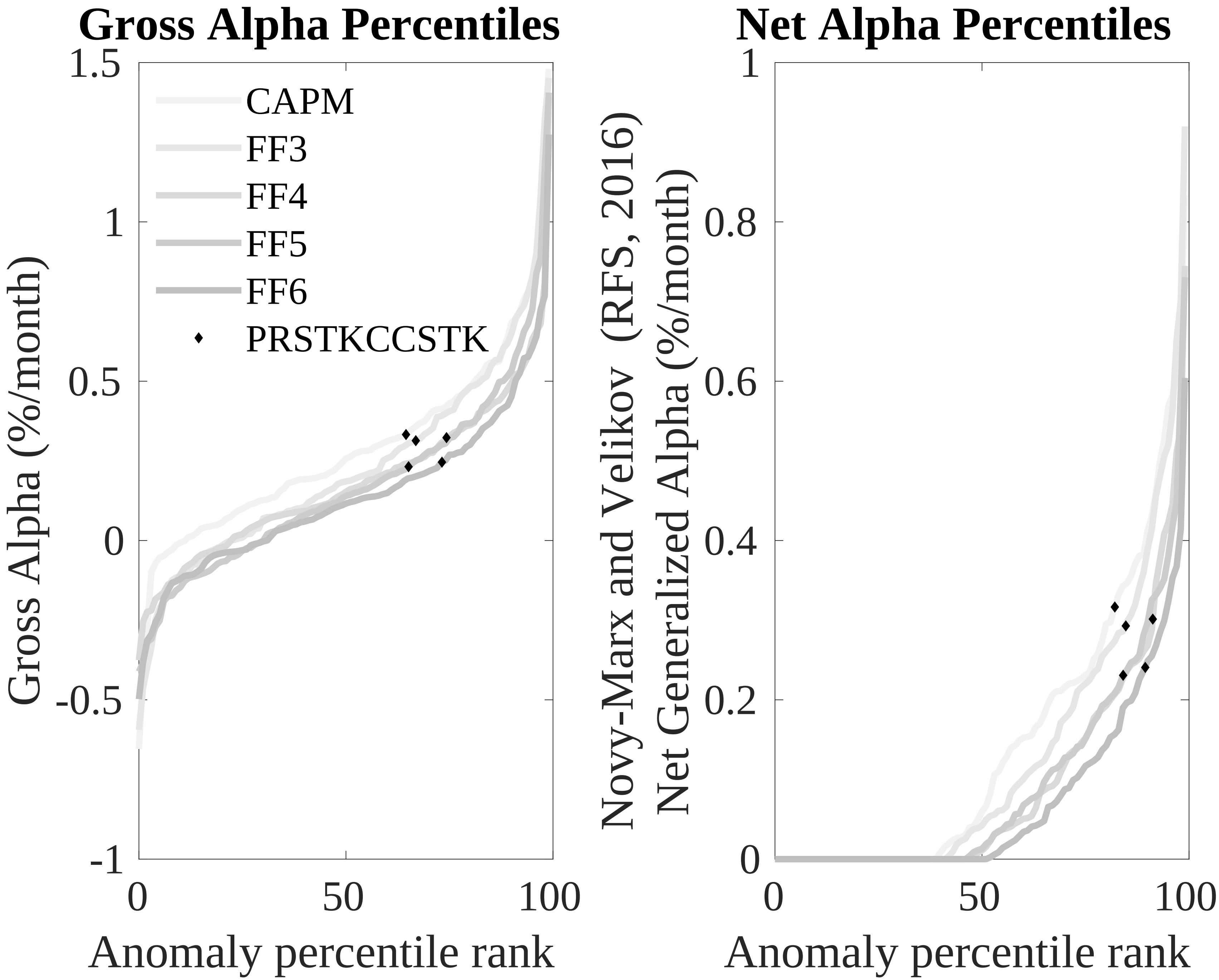
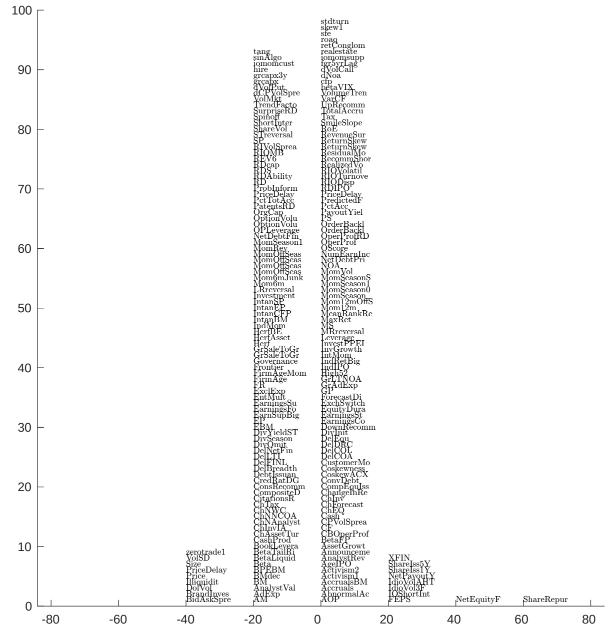
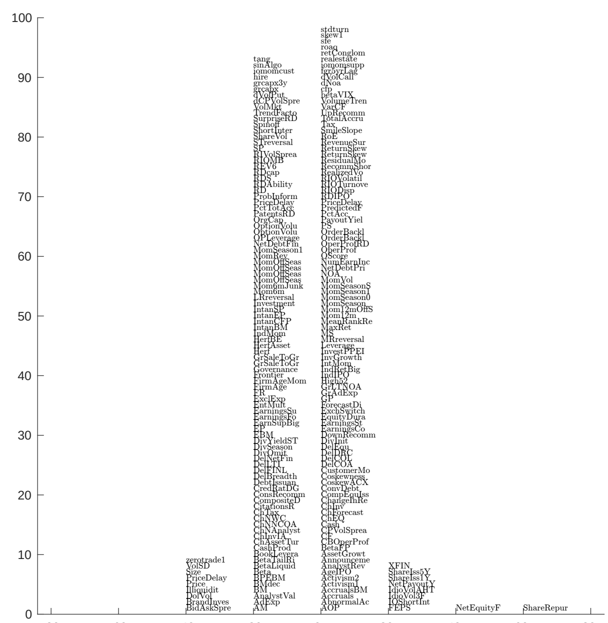


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
PRSTKCCSTK
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank



Panel A: Pearson



Panel B: Spearman

Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

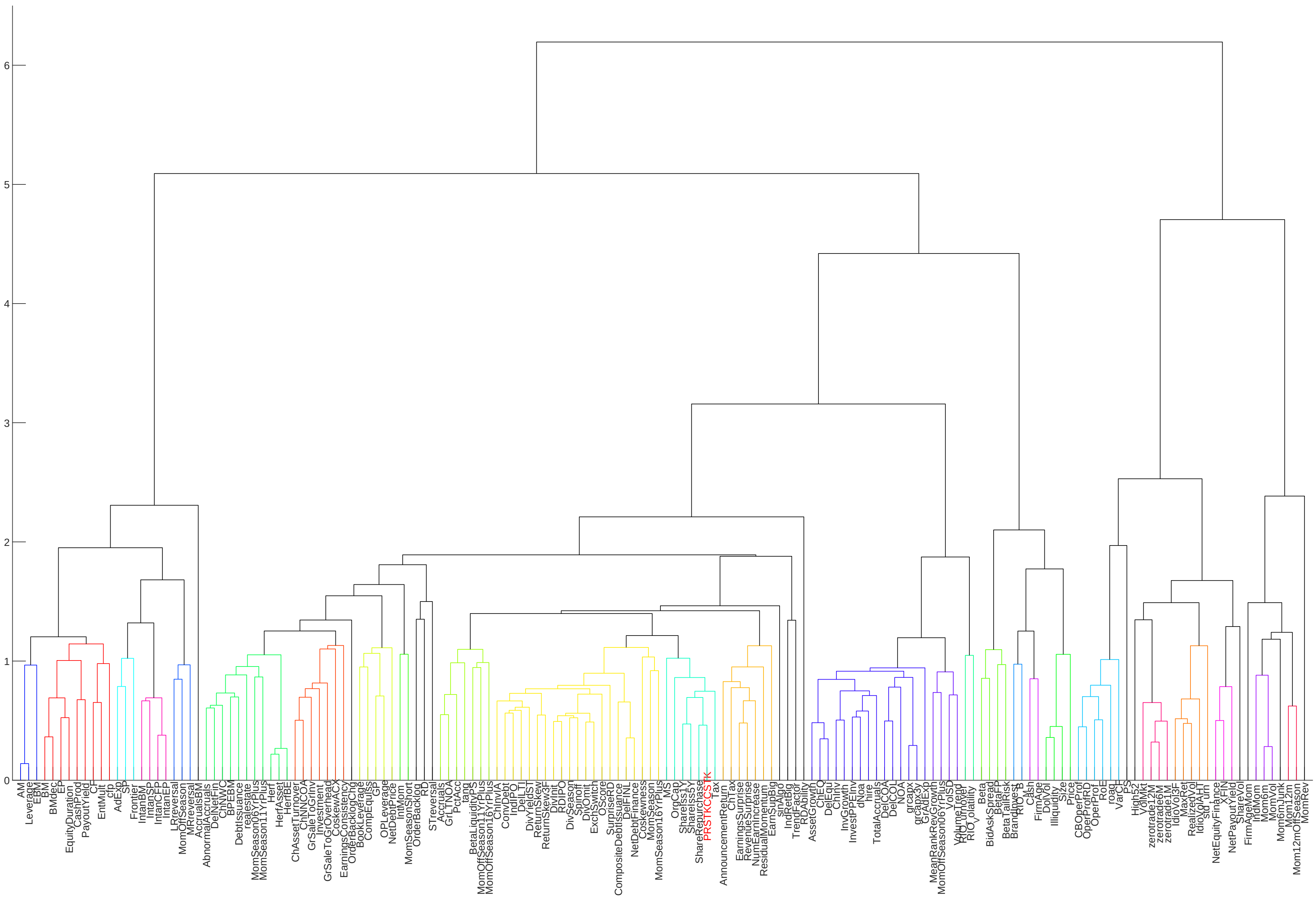


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

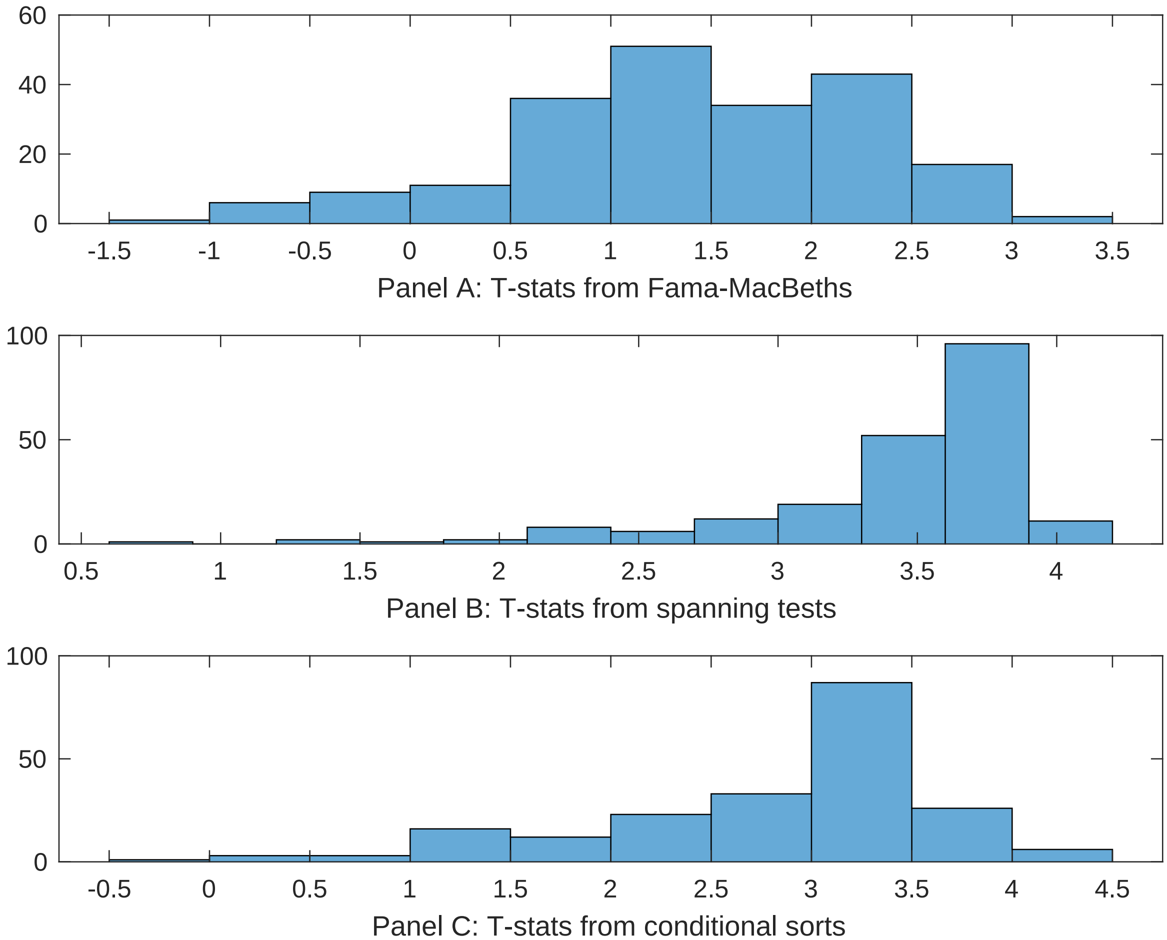


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI} EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EBI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBI} EBI_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.10	0.12	0.13	0.12	0.13	0.13	0.13
	[3.85]	[5.25]	[5.40]	[5.15]	[5.32]	[5.96]	[5.80]
EBI	-0.64	0.11	0.13	1.00	0.68	0.24	0.73
	[-0.06]	[0.96]	[1.44]	[0.82]	[0.80]	[2.23]	[0.53]
Anomaly 1	0.24						0.54
	[2.67]						[1.09]
Anomaly 2		0.17					-0.28
		[2.79]					[-3.35]
Anomaly 3			0.17				0.63
			[8.44]				[2.89]
Anomaly 4				0.28			0.12
				[5.04]			[2.36]
Anomaly 5					0.19		0.14
					[7.18]		[5.99]
Anomaly 6						0.32	0.26
						[6.99]	[5.68]
# months	624	624	619	619	624	619	619
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EBI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.29	0.28	0.23	0.26	0.23	0.18	0.27
	[4.23]	[3.67]	[2.99]	[3.19]	[2.87]	[2.23]	[4.05]
Anomaly 1	63.95						52.36
	[18.12]						[13.38]
Anomaly 2		40.10					17.78
		[11.70]					[4.03]
Anomaly 3			39.35				10.80
			[10.02]				[2.45]
Anomaly 4				18.42			-6.10
				[5.71]			[-1.90]
Anomaly 5					28.79		5.29
					[7.08]		[1.34]
Anomaly 6						27.23	-3.17
						[6.15]	[-0.74]
mkt	3.25	9.25	3.05	5.36	6.45	3.26	5.89
	[2.06]	[4.97]	[1.70]	[2.72]	[3.29]	[1.73]	[3.56]
smb	-7.00	-1.36	-15.13	-13.30	-7.93	-15.28	-1.04
	[-2.89]	[-0.46]	[-5.61]	[-4.59]	[-2.56]	[-5.41]	[-0.39]
hml	-6.00	-2.92	-0.11	-6.45	4.24	2.86	-4.15
	[-1.97]	[-0.86]	[-0.03]	[-1.68]	[1.17]	[0.79]	[-1.30]
rmw	-3.05	-7.69	-7.06	4.68	-0.31	5.29	-15.37
	[-0.93]	[-1.91]	[-1.68]	[1.11]	[-0.07]	[1.30]	[-3.92]
cma	-3.67	-13.79	-11.74	-3.37	-8.43	-0.61	-14.32
	[-0.79]	[-2.50]	[-2.06]	[-0.56]	[-1.40]	[-0.11]	[-2.74]
umd	-2.42	-3.04	-4.06	-1.15	-2.54	-3.78	-3.46
	[-1.55]	[-1.74]	[-2.27]	[-0.61]	[-1.37]	[-2.02]	[-2.26]
# months	624	624	620	620	624	620	620
$\bar{R}^2(\%)$	43	28	25	17	19	17	47

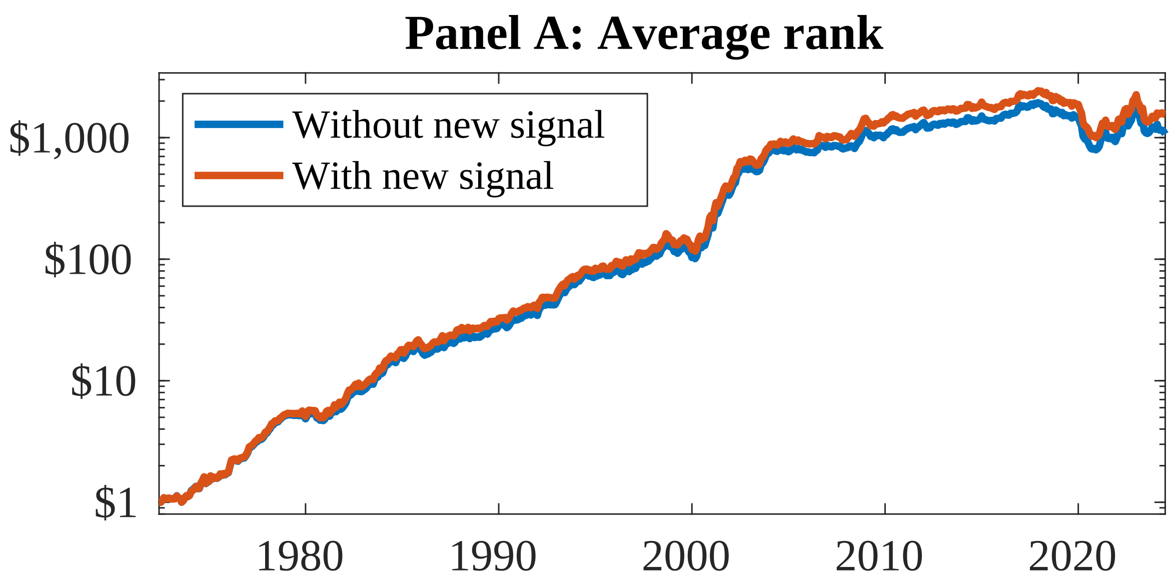


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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